Error Analysis — Reading Traces Like a Professional

Build Week

Turn "it fails sometimes" into a ranked list of real problems, found by reading your app's actual answers.

At a glance

Module	Week 5 · Module 3 — Evals & Error Analysis (The Core)
Format	Build week — hands-on, one deliverable
Deliverable	≈20 traces read, open-coded notes, named problem groups, ranked taxonomy
Mentor review	Friday — informal, no marks
Evaluated	Week 6 · Monday

What this week is about

Until now you fixed whatever problems you happened to notice. That misses the ones you didn't see. This week you become systematic: read a real batch of the app's answers, honestly note what went wrong in each, and turn that into a ranked list of the top problems worth fixing.

Why it matters

Why read the answers by hand — can't a tool do it?	What is a "trace"?	Why rank the problems?
Reading your own failures is the one part that can't be automated. It's where you discover problems you never guessed were there.	A full record of one request — the question, what the app fetched, and what it answered — complete enough to replay later.	You can't fix everything at once. Ranking by how often and how badly each one hurts tells you what to fix first.

What you'll learn

- What a complete "trace" is — enough detail to replay any answer later
- Taking a fair, random sample instead of cherry-picking nice examples
- Reading each failure and writing one honest sentence about what went wrong — before deciding on categories
- Grouping those notes into a handful of named problem types
- Ranking the problem types by how often they happen and how much they hurt
- Picking the one problem to attack next, and writing down what you expect to happen

Topics covered

The exact concepts to study.

• Complete traces	• Random vs curated sampling
• Open coding	• Error taxonomy
• Frequency × severity	• Choosing the fix target
• Benchmarks (MMLU, HumanEval) vs your app	• Writing a prediction first

Your task this week

Evaluated: Week 6 · Monday

Collect around 20 real answers from your app, read every one, and write a short honest note on each failure. Group the notes into a few named problems and rank them, so you have a clear list of what to fix first.

Assigned tracks

Track	Topic	Deliverable
A	Customer support tickets	Read 20 real traces and hand back a ranked taxonomy
B	Recipes & food	Read 20 real traces and hand back a ranked taxonomy
C	HR policy	Read 20 real traces and hand back a ranked taxonomy
D	Insurance claims	Read 20 real traces and hand back a ranked taxonomy
E	Developer documentation	Read 20 real traces and hand back a ranked taxonomy
F	Legal contracts	Read 20 real traces and hand back a ranked taxonomy

Same task, six different topics — so everyone works on their own version. Download whichever you are assigned.

How your mentor checks

Friday · no marks

A quick review of your task. Not a test — just a few things your mentor looks at, so you can fix anything before Monday.

- Did they read a fair sample, not just the good examples?
- Is there an honest note per failure, written before grouping them?
- Do the problem groups have clear names a stranger would understand?
- Are the problems ranked, with a chosen target to fix next?